SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1004	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:		**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

*I **M.DEVI SRILATHA (192525236)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ***Srilatha***

Annexure A – Architecture Design Assignment

Instructions to Students:

Each student will be assigned an **individual software project problem statement**. Based on the assigned problem, analyze the system requirements and design an appropriate software architecture by applying software engineering principles. Your submission should demonstrate your ability to select, justify, and evaluate a suitable architectural style.

Question

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

1. Analyze System Requirements

- Study the given problem statement and identify the system objectives.
- Analyze the functional and non-functional requirements that influence the software architecture.
- Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

2. Select a Suitable Software Architecture

- Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).
- Justify why the selected architecture is the most suitable for the given problem.

3. Draw the Software Architecture Diagram

- Design a clear architecture diagram illustrating the overall structure of the proposed system.
- Include all major layers/components, databases, external systems, APIs, and communication mechanisms.
- Use appropriate software architecture notations and labels.

4. Explain Components and Their Interactions

- Describe the purpose and responsibilities of each architectural component.
- Explain how the components communicate and exchange data.
- Illustrate the overall workflow of the system.

5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

- Scalability
- Security
- Performance
- Reliability
- Maintainability
- Availability

6. Justify Architectural Decisions

- Explain the rationale behind your architectural choices.
- Discuss the trade-offs considered while selecting the architecture.
- Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and	Diagram is incomplete, unclear, or

	major components, interfaces, and interactions accurately represented.		lacks sufficient detail.	technically incorrect.
Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation (10%)	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

Criteria	**Mark**
System Requirement Analysis **(30%)**	/5
Selection of Software Architecture **(25%)**	/5
Architecture Diagram and Component Design **(20%)**	/5
Component Interaction and Quality Attribute Analysis **(15%)**	/5
Architectural Justification and Technical Presentation **(10%)**	/5
TOTAL	/25

Question:

Design and develop an AI-Based Legal Document Management System that securely stores legal documents, classifies documents, supports intelligent search, tracks versions, and provides reminders for important dates and contractual obligations.

1. Introduction

The AI-Based Legal Document Management System is a software application designed to help lawyers, legal assistants, administrators, and clients manage legal documents efficiently.

The system provides secure document storage, document retrieval, AI-based document classification, OCR text extraction, intelligent search, case management, deadline tracking, and notifications.

The main objective is to reduce manual paperwork, improve document organization, protect confidential legal information, and make legal document management faster and easier.

2. System Requirements and Quality Attributes

2.1 Functional Requirements

1. **User Authentication:** The system verifies user login credentials and provides role-based access.
2. **Document Management:** Users can upload, store, retrieve, and manage legal documents.
3. **OCR Text Extraction:** The system extracts text from scanned legal documents.
4. **Document Classification:** The system automatically categorizes documents based on their content.
5. **Intelligent Search:** Users can search for documents using keywords and extracted text.
6. **Legal Case Management:** Users can maintain case details and track important deadlines.
7. **Notification Service:** The system sends email or SMS notifications about upcoming deadlines.
8. **Database Management:** The system stores user details, document information, case records, and audit logs.
9. **Backup and Recovery:** The system supports data backup and recovery to reduce the risk of data loss.
10. **Audit Logging:** The system records important user activities for monitoring and accountability.

2.2 Quality Attributes

- **Security:** Protects confidential legal documents through authentication and access control.
- **Performance:** Enables faster document searching and retrieval.
- **Scalability:** Supports the addition of more users and documents as the system grows.
- **Reliability:** Maintains consistent operation and reduces the risk of data loss.
- **Maintainability:** Allows individual components to be updated without redesigning the entire system.
- **Availability:** Helps authorized users access the system when required, subject to service availability.

3. Selection of Software Architecture Style

The proposed system uses a **Component-Based Software Architecture**.

In this approach, the application is divided into separate, connected software components. Each component performs a specific function, such as document management, user authentication, intelligent search, or notification handling.

The components communicate with one another to complete the system's overall operations.

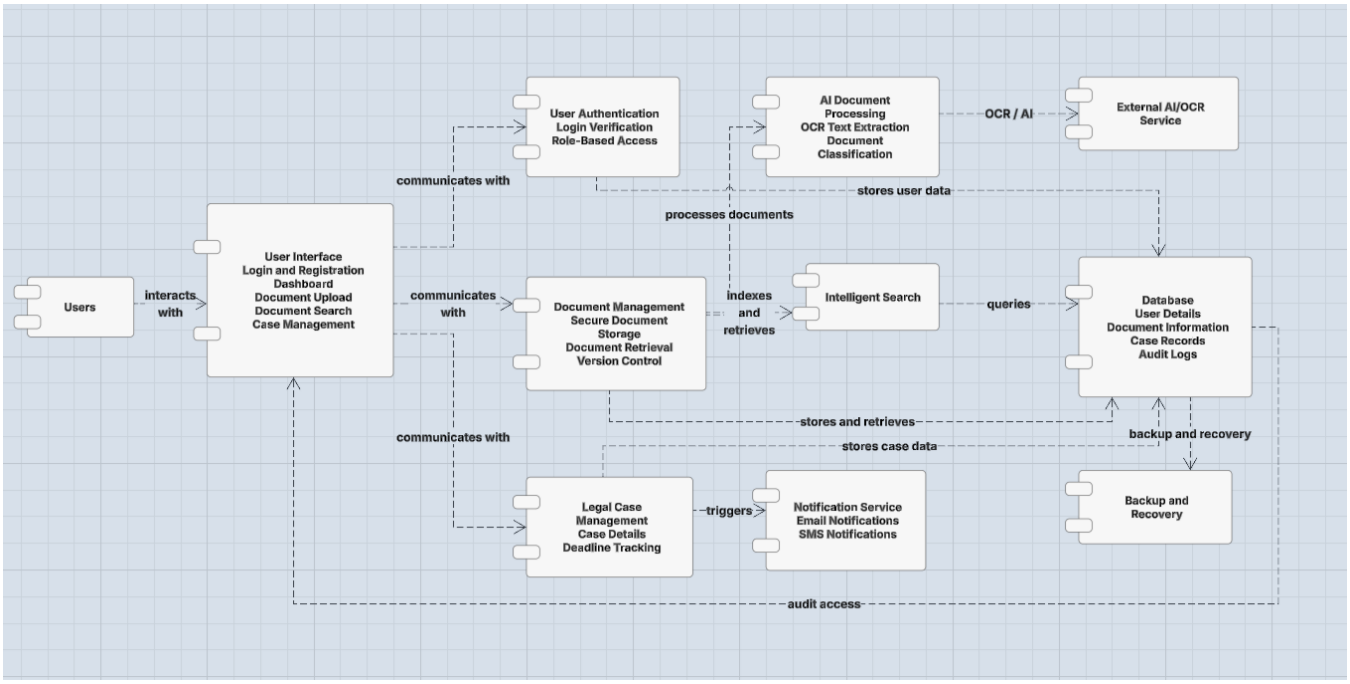
Justification

1. **Modularity:** Each component has a clearly defined responsibility.
2. **Maintainability:** Individual components can be modified or improved more easily.
3. **Reusability:** Components such as authentication and notification services may be reused in other applications.
4. **Scalability:** Components can be enhanced as the number of users and documents increases.
5. **Integration:** AI/OCR services and backup services can be connected to the main application.
6. **Testing:** Individual components can be tested separately before testing the complete system.

This architecture is suitable because the project contains several related functions that must work together as one integrated software application.

4. Software Architecture Diagram

The architecture diagram was created using **Umbrello UML Modeller**.



Components Represented in the Diagram

The diagram contains the following components:

1. Users
2. User Interface
3. User Authentication
4. Document Management
5. AI Document Processing
6. Intelligent Search
7. Legal Case Management
8. Notification Service
9. Database
10. External AI/OCR Service
11. Backup and Recovery

The arrows represent the communication and interactions between these components.

5. Description of Components

5.1 Users

Users include lawyers, legal assistants, administrators, and clients. They interact with the system to manage documents and legal cases.

5.2 User Interface

The User Interface provides login and registration, a dashboard, document upload, document search, and case management options. It allows users to access the system's features.

5.3 User Authentication

This component verifies login credentials and controls access according to user roles. It helps prevent unauthorized access to confidential information.

5.4 Document Management

This component handles secure document storage, document retrieval, and version control. It helps users organize and manage legal documents efficiently.

5.5 AI Document Processing

This component uses AI-based techniques to classify legal documents and organize them according to their content.

It also supports OCR text extraction, which converts text in scanned documents or images into machine-readable text.

5.6 Intelligent Search

This component helps users locate documents using keywords and extracted text. It reduces the time required to find specific legal information.

5.7 Legal Case Management

This component maintains legal case details and tracks important deadlines. It helps users organize case-related information and monitor upcoming tasks.

5.8 Notification Service

This component sends email or SMS notifications about important case deadlines and reminders. It helps users avoid missing scheduled activities.

5.9 Database

The database stores user details, document information, case records, and audit logs. It provides the information required by different system components.

5.10 External AI/OCR Service

This external service supports document text extraction and AI-based processing. The main application communicates with it to process uploaded documents when required.

5.11 Backup and Recovery

This component supports the backup of important system information and helps restore data when necessary. Backup copies must be protected against unauthorized access.

6. Component Communication and Workflow

The system follows the workflow described below.

Step 1: User Access

The user opens the application through the User Interface and enters their login credentials.

Step 2: Authentication

The User Authentication component verifies the credentials and checks the user's access permissions.

Step 3: Document Upload

After successful authentication, the user uploads a legal document through the User Interface.

Step 4: Document Processing

The Document Management component handles the uploaded document. The AI Document Processing component performs OCR text extraction and document classification when required.

Step 5: Document Storage

The document and its related information are stored securely. The database maintains document metadata and other relevant records.

Step 6: Intelligent Search

When a user searches for a document, the Intelligent Search component searches the available document information and extracted text to identify relevant results.

Step 7: Case Management

Users maintain case details and deadlines through the Legal Case Management component.

Step 8: Notifications

When a deadline requires a reminder, the Notification Service can send an email or SMS notification through the configured external service.

Step 9: Backup and Recovery

The system performs scheduled or configured backups and supports data restoration when necessary.

Step 10: Audit Logging

Important activities, such as document uploads and access-related events, are recorded in audit logs according to the system's security rules.

7. Communication Mechanisms

The components communicate using the following mechanisms:

Communication	Purpose
User Interaction	Allows users to operate the application through the interface.
HTTPS	Protects communication between the user's browser and the application.
REST API	Allows application components to exchange requests and responses.
Database Queries	Store, retrieve, and update system information.
AI/OCR API	Sends documents or processing requests to the external AI/OCR service and receives results.
Notification API	Sends requests to an email or SMS service.
Backup and Recovery Operations	Creates backup copies and supports data restoration.

These are the proposed communication mechanisms. Their exact implementation will depend on the technologies selected during development.

8. Quality Attribute Analysis

8.1 Security

Security is important because legal documents may contain confidential information. The system uses authentication and role-based access control to restrict access. HTTPS, secure document storage, protected backups, and audit logging further support security.

8.2 Performance

Intelligent search and OCR-based text extraction help users locate relevant documents faster. Search indexing and appropriate database queries can improve performance when the document collection grows.

8.3 Scalability

The system can be extended by adding more users, documents, and case records. Individual components can also be optimized as demand increases.

8.4 Reliability

Database protection, error handling, backups, and recovery procedures help reduce the impact of failures. Regular testing is necessary to verify that documents can be recovered correctly.

8.5 Maintainability

The component-based design makes it easier to update features such as OCR processing, authentication, or notifications without redesigning the entire application.

8.6 Availability

The application should be accessible to authorized users whenever required. Availability depends on reliable hosting, network connectivity, database operation, and recovery procedures.

9. Design Decisions, Trade-Offs, and Future Enhancements

9.1 Design Decisions

- Component-based architecture was selected to organize the system into manageable software components.
- A database is used to manage structured information, including users, document metadata, and case records.
- An external AI/OCR service is included to support text extraction and document processing.
- A separate notification component handles reminders and deadline alerts.
- Backup and recovery are included to help protect important information.

9.2 Trade-Offs

- **Separate components:** They improve maintainability but require clear communication interfaces and additional integration testing.
- **OCR processing:** It makes scanned documents searchable, but extraction accuracy can decrease when scans are unclear or poorly formatted.
- **Notifications:** Email and SMS reminders are useful, but delivery depends on the availability of the external notification provider.
- **Backup and recovery:** Backups improve resilience but require additional storage, security controls, and regular recovery testing.

9.3 Future Enhancements

1. Add multilingual document processing.
2. Introduce digital signatures and document verification.
3. Improve AI-based legal document summarization.
4. Add advanced semantic search.
5. Implement multi-factor authentication.
6. Provide a mobile-friendly interface.
7. Add secure document sharing and approval workflows.
8. Introduce automated backup monitoring and recovery testing.

10. Conclusion

The AI-Based Legal Document Management System provides an organized approach to managing legal documents and case information. The component-based architecture connects user authentication, document management, AI processing, intelligent search, case management, notifications, database storage, and backup facilities.

The architecture improves modularity and maintainability while providing a foundation for security, performance, scalability, and reliability. The proposed design can be extended with additional AI capabilities and legal workflow features as the project develops.